COMEX COPPER FUTURES (HG)

Spread & Curve Structure — Fundamental Reference

Mine Supply, Chinese Demand, the Energy Transition, and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of COMEX Copper futures (ticker: HG), contract size 25,000 pounds, with particular focus on the forces that shape the forward curve and the spread structure between contract months. Copper is widely regarded as the commodity most representative of global industrial activity — earning it the nickname "Dr. Copper" for its perceived diagnostic value regarding the health of the world economy. Its fundamentals combine the long-cycle characteristics of a capital-intensive mined commodity (multi-year mine development lead times, geographically concentrated production in Chile and Peru) with the demand-side dominance of a single country, China, which consumes more than half of global refined copper. Copper additionally sits at the centre of the energy transition narrative — electric vehicles, renewable power generation, and grid infrastructure all require substantially more copper per unit of output than the technologies they replace. Understanding mine supply economics, the Chinese demand cycle, exchange inventory dynamics, and the global trade architecture spanning COMEX, the London Metal Exchange (LME), and the Shanghai Futures Exchange (SHFE) is the foundation of spread and butterfly trading on the HG curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and the Three-Exchange Market

COMEX Copper is one leg of a globally interconnected three-exchange copper market. Understanding the relationship between COMEX, the London Metal Exchange (LME), and the Shanghai Futures Exchange (SHFE) is essential before considering any single exchange's fundamentals in isolation.

Parameter	Detail
Exchange	CME Group — COMEX division (New York)
Ticker	HG (front month)
Contract size	25,000 pounds (approximately 11.34 metric tonnes)
Quotation	US cents per pound (¢/lb); minimum tick = 0.0005¢/lb = $12.50 per contract
Contract months	All 12 calendar months; active contracts extend approximately 2–3 years forward. Liquid trading concentrated in front 6–12 months.
Settlement type	Physical delivery of Grade 1 Electrolytic Copper Cathode (min. 99.99% purity, LME/COMEX-approved brands) at COMEX-licensed warehouses in the US (and select international locations).
Trading hours	CME Globex: Sunday–Friday 18:00–17:00 ET (23-hour session); most liquid 08:00–13:00 ET (overlap with LME)
Last trading day	Third-to-last business day of the delivery month
Contract value at $4.50/lb	$112,500 per contract
Other major venues	LME (London Metal Exchange) — the global reference price, quoted in USD/metric tonne, with daily and 3-month forward contracts plus a full forward curve out to ~10 years (less liquid beyond 15 months). SHFE (Shanghai Futures Exchange) — the Chinese domestic benchmark, quoted in CNY/tonne, primarily reflecting onshore Chinese supply/demand.
COMEX-LME basis	COMEX HG converted to $/tonne is typically compared to LME 3-month. The COMEX-LME spread (the "arb") reflects US domestic premium/discount, freight, and the relative tightness of US vs global inventories.

1.1 The Three-Exchange Structure — Why It Matters

- **LME as the global price reference**: the LME has traded copper since 1877 and remains the primary global price discovery venue. LME copper has a unique daily prompt date structure (any business day can be a delivery date for the first 3 months), creating an extremely granular forward curve unlike the monthly-contract structure of COMEX. LME warehousing spans Asia, Europe, and the Americas, making LME stock levels a genuinely global supply indicator.
- **COMEX as the US domestic reference and the most liquid US-hours venue**: COMEX HG futures are the primary instrument for US-based funds, industrials, and producers to manage copper price risk. COMEX pricing tends to carry a premium over LME reflecting the cost of importing refined copper into the US (the US is a net copper importer) — this premium structure became dramatically more important after 2024-2025 tariff developments (see Section 8).
- **SHFE as the Chinese domestic reference**: China consumes more than half of global refined copper, and SHFE prices reflect onshore Chinese conditions, including capital controls that partially insulate the Chinese price from instantaneous global arbitrage. The SHFE-LME arbitrage (adjusted for VAT, duties, and freight) is a key signal of Chinese import incentives: when the arb is open (SHFE high enough relative to LME plus costs), China imports more refined copper, tightening the global market.

- **Inter-exchange stock arbitrage**: traders can in principle move warehouse stock between LME, COMEX, and (with more friction due to Chinese capital controls) SHFE when price differentials exceed transport and financing costs. This trade flow is a major driver of reported exchange inventory swings and is often mistaken for a change in physical consumption when it is actually a relocation of existing stock between warehousing systems.

2. Mine Supply — Geographic Concentration and Project Economics

World mined copper production is approximately 22–23 million tonnes per year. Unlike many agricultural commodities, copper supply is geographically concentrated in a small number of countries and is characterised by extremely long project lead times — a defining structural feature that makes the supply side slow to respond to price signals.

Country	Mined Production (approx.)	Key Districts/Mines	Structural Risk	HG Curve Impact
Chile	5.2–5.5 Mt	Escondida (BHP/Rio Tinto), Collahuasi, Chuquicamata, El Teniente (Codelco)	Water scarcity in the Atacama (driving desalination capex); declining ore grades at legacy mines; labour strikes (historically frequent at Escondida); permitting delays for new projects	Single most important country for global supply. Chilean strike action or major mine disruption is the most reliable acute bullish catalyst in copper.
Peru	2.6–2.8 Mt	Cerro Verde, Antamina, Las Bambas (MMG), Cuajone	Social/community conflict and road blockades (Las Bambas has faced repeated transport route blockades); political instability; permitting	Second-largest producer. Community blockade disruptions at Las Bambas have repeatedly removed material tonnage from the export pipeline with limited advance warning.
China	1.7–1.9 Mt mined; much larger refined	Jiangxi, Yunnan, Tibet (growing)	Domestic mined ore is a small share of Chinese refined output — China imports the majority of its concentrate and refined cathode needs	China's mined output is modest relative to its consumption; the country's role is overwhelmingly on the demand and refining side, not primary mine supply.
DR Congo	2.8–3.0 Mt	Kamoa-Kakula (Ivanhoe/Zijin), Tenke Fungurume, Kamoto	Fastest-growing major supply source; political/governance risk; infrastructure and power constraints; cobalt by-product co-dependency	The most important new supply growth region of the past decade. Kamoa-Kakula expansions are closely tracked as the marginal source of new global supply.
United States	1.1–1.3 Mt	Morenci, Bingham Canyon/Kennecott, Bagdad (Arizona, Utah dominant)	Permitting timelines (Resolution Copper in Arizona delayed for years by legal/regulatory disputes); declining grades at mature operations	Directly relevant to COMEX deliverable supply and to the domestic premium discussion (Section 8).
Indonesia	0.9–1.1 Mt	Grasberg (Freeport-McMoRan)	One of the world's largest ore bodies but transitioning from open-pit to underground mining, which carries execution risk; export permit policy	Underground transition at Grasberg is a multi-year supply variable closely tracked by analysts.
Australia, Russia, Mexico, Zambia, Kazakhstan (other major)	~3.5 Mt combined	Various	Variable — sanctions risk (Russia), power/water (Zambia)	Collectively significant; Zambian and Russian supply face distinct policy risk profiles.

2.1 Declining Ore Grades — The Structural Cost Treadmill

- **Ore grade decline**: the average grade of copper ore being mined globally has declined steadily for decades as higher-grade deposits are depleted and miners increasingly process lower-grade ore. Global average head grades have fallen from approximately 1.0–1.2% copper content several decades ago to approximately 0.5–0.6% today at many major operations. Lower grade means more ore (and more energy, water, and waste rock) must be processed to produce the same quantity of metal — a structural cost increase embedded in the supply curve.
- **Strip ratio increases**: as open-pit mines deepen over their operating life, the ratio of waste rock to ore removed (strip ratio) rises, increasing the cost per tonne of contained copper. This is a predictable, mine-specific trajectory that analysts model explicitly for major operations.

2.2 Mine Development Lead Times — The Defining Supply Constraint

- **From discovery to first production: typically 10–20 years**. This includes exploration and resource definition (years), feasibility studies, permitting (often the single largest source of delay, particularly in jurisdictions with extensive environmental review and indigenous/community consultation requirements), financing, and construction (typically 3–5 years for a major project). This extraordinarily long lead time means that even a sustained, multi-year period of high copper prices cannot produce a meaningful supply response for the better part of a decade.
- **Permitting risk as a quantifiable delay factor**: Resolution Copper (Arizona, Rio Tinto/BHP) — one of the largest undeveloped copper resources in North America — has been in the permitting process for more than a decade amid litigation over the project's location on land sacred to the San Carlos Apache Tribe. Pebble Mine (Alaska) was effectively blocked by a US EPA Clean Water Act determination. These cases illustrate that permitting timelines, not geology or capital availability, are frequently the binding constraint on new supply in developed-market jurisdictions.
- **Capital intensity**: major new copper mines require capital expenditure often exceeding $5–10 billion for large-scale projects. This capital intensity means project sanctioning decisions require a board-level commitment to a multi-decade price view, making major mining companies cautious about committing capital based on short-term price spikes.

The combination of declining grades and 10-20 year development lead times means copper mine supply is fundamentally inelastic over the 1-5 year horizon relevant to most spread trading. Price spikes are resolved primarily through demand destruction or substitution, not through a rapid mine supply response — a structural feature shared with gold but absent in agricultural and shale-driven energy markets.

3. Refining, Concentrate, and Scrap — The Supply Chain

Mined copper ore must be processed through several stages before it reaches the refined cathode form that is deliverable against COMEX, LME, and SHFE contracts. Understanding this processing chain — and where China's dominance is concentrated within it — is essential for interpreting supply data correctly.

3.1 From Ore to Cathode

- **Concentrate route (the majority of supply)**: mined ore is crushed and processed at the mine site via flotation to produce copper concentrate (typically 20-30% copper content). Concentrate is then shipped — often internationally — to smelters, which produce blister or anode copper, and then to refineries, which produce 99.99% pure cathode via electrolytic refining.
- **SX-EW route (solvent extraction-electrowinning)**: lower-grade oxide ores can be processed via heap leaching and SX-EW to produce cathode copper directly at the mine site, bypassing the smelting stage entirely. This route is significant in Chile and represents a growing share of supply as sulphide ore grades decline and oxide/leachable ore processing becomes more economic.
- **China's smelting dominance**: China has built, by a wide margin, the world's largest copper smelting and refining capacity, processing both domestic and (overwhelmingly) imported concentrate. Chinese smelters process roughly half of global mined concentrate, making China the indispensable midstream processor even though its own mined output is comparatively modest.

3.2 Treatment and Refining Charges (TC/RCs) — The Smelter Margin Signal

- **What TC/RCs represent**: mining companies sell concentrate to smelters and pay a Treatment Charge (TC, per tonne of concentrate) and a Refining Charge (RC, per pound of contained copper) to compensate the smelter for processing. TC/RCs are negotiated annually (historically anchored by benchmark settlements between major miners like Freeport or Antofagasta and major Chinese smelters such as Jiangxi Copper) and also traded on a spot basis throughout the year.
- **TC/RCs as a real-time concentrate market tightness signal**: when global smelting capacity exceeds available concentrate supply (a "concentrate-short" market), smelters compete for limited concentrate and TC/RCs fall — in extreme cases turning negative, meaning smelters effectively pay miners to take the concentrate, an unprecedented condition that emerged in 2023-2024 as massive Chinese smelting capacity additions outpaced concentrate mine supply growth. This is one of the most informative leading indicators of the physical mine-to-refinery supply balance, though it operates with a different transmission mechanism than the refined cathode price itself.
- **Smelter margin squeeze and curtailment risk**: when TC/RCs are severely compressed or negative, smelter profitability is impaired, raising the prospect of smelter production curtailments — particularly among higher-cost, smaller Chinese smelters. Curtailments reduce refined cathode supply even when mined concentrate supply is unaffected, a distinct transmission channel from a mine-level disruption.

3.3 Scrap Copper — The Secondary Supply Source

- **Scrap as a significant supply component**: recycled (secondary) copper accounts for approximately 30-35% of total global copper supply when counting both "old scrap" (end-of-life products: wiring, motors, demolished buildings) and "new scrap" (manufacturing offcuts recycled within the supply chain). Secondary refined production requires substantially less energy than primary (ore-based) production.
- **Scrap availability and price elasticity**: scrap collection and processing responds to price with a much shorter lag than mine supply — higher copper prices incentivise increased scrap collection and processing within weeks to months, providing a faster-responding (though smaller in absolute terms) supply buffer than primary mine production.

- **Chinese scrap import policy**: China's 2018-2021 restrictions on scrap imports (part of a broader "National Sword" policy targeting waste imports) significantly altered global scrap trade flows, redirecting scrap processing capacity investment toward Malaysia, Vietnam, and other Southeast Asian countries, and increasing China's reliance on refined cathode imports and domestic mine/smelter output to compensate.

4. China — The Dominant Demand Variable

China consumes more than half of the world's refined copper — a concentration of demand in a single country that has no parallel among the other base metals and exceeds even oil or natural gas in degree. Chinese economic data, policy stimulus decisions, and sector-specific demand drivers are therefore the single most important demand-side variable for HG futures.

4.1 Chinese End-Use Demand Sectors

Sector	Share of Chinese Copper Demand	Key Driver	Data Signal
Power grid and electrical infrastructure	~45–50%	State Grid Corporation capex; ultra-high-voltage (UHV) transmission line construction; renewable energy grid connection buildout	State Grid annual/monthly capex announcements; UHV project approvals; NDRC grid investment plans
Construction and real estate	~15–20%	Wiring, plumbing, air conditioning in new and renovated buildings; property sector investment and completions	New home starts/completions; property investment YoY; this has been a structural headwind since the 2021+ property sector deleveraging
Consumer durables / appliances (white goods)	~10–12%	Air conditioner, refrigerator production (each uses substantial copper tubing and wiring); domestic and export appliance demand	China Household Electrical Appliances Association production data; export order books
Transportation (incl. EVs)	~10–12%	Traditional auto wiring harnesses; electric vehicle production (EVs use roughly 3-4x the copper of an internal combustion vehicle)	China Association of Automobile Manufacturers (CAAM) monthly EV production/sales data
Industrial machinery / electronics	~10%	General manufacturing capex; electronics and semiconductor-adjacent production	Manufacturing PMI; industrial production data

4.2 The State Grid Corporation of China — The Single Largest Buyer

- **State Grid Corporation of China (SGCC)** is the largest single copper consumer in the world, responsible for the operation and expansion of the majority of China's electricity transmission and distribution network. SGCC publishes annual capital expenditure plans (typically announced in Q1) that are closely tracked as the single most important demand-side data point in the global copper market.
- **Ultra-high-voltage (UHV) transmission**: China has invested heavily in UHV transmission lines to move electricity (increasingly from renewable generation in the west and north) to demand centres on the east coast. UHV lines are exceptionally copper-intensive. New UHV project approvals and construction milestones are tracked as leading indicators of grid-related copper demand roughly 1-2 years ahead of physical consumption.
- **Renewable grid connection**: the build-out of solar and wind generation capacity in China requires substantial copper for grid interconnection, inverters, and cabling — making renewable capacity addition targets (published in China's Five-Year Plans and annual energy targets) a structural demand driver layered on top of base grid maintenance capex.

4.3 The Chinese Property Sector — A Structural Headwind Since 2021

- **Pre-2021 role**: property construction and related wiring, plumbing, and HVAC installation was historically one of the largest single components of Chinese copper demand, closely tracking the property investment boom of the 2000s-2010s.
- **The 2021+ deleveraging**: the "three red lines" policy constraining developer leverage, the Evergrande default and subsequent broader developer distress, and a multi-year decline in new home starts have made the property sector a persistent demand headwind for Chinese copper consumption since 2021, partially offset by continued strength in grid and EV-related demand. Property data (new starts, floor space under construction, completions) remains closely tracked specifically for its (now reduced but still relevant) copper demand implications.

4.4 Chinese Macro Stimulus and Copper

- **Infrastructure stimulus packages**: Chinese fiscal and monetary stimulus announced in response to growth slowdowns (reserve requirement ratio cuts, special local government bond issuance for infrastructure, targeted property sector support measures) are closely watched events for copper, since infrastructure-heavy stimulus has historically been copper-intensive in its transmission to the real economy.
- **PMI and industrial production data**: China's official and Caixin Manufacturing PMI readings are the most frequently cited leading indicators for near-term Chinese industrial copper demand, published monthly and watched closely by copper traders globally.
- **State Reserve Bureau (SRB) stockpiling and releases**: China's State Reserve Bureau holds and periodically buys or releases strategic copper reserves. SRB purchases (often during price dips, serving a dual function of price support and strategic stockpiling) and releases (during periods of perceived domestic tightness or as a price-moderating tool) are not transparently disclosed in real time but are inferred by analysts from trade flow and SHFE/bonded warehouse stock data, making SRB activity an important but imperfectly observable variable.

5. The Energy Transition — Structural Demand Growth

Copper's role in electrification has made it one of the most closely watched commodities in the energy transition narrative. Electric vehicles, renewable power generation, and grid modernisation all require substantially more copper per unit of output than the technologies they displace, creating a structural demand growth thesis that operates on a multi-year horizon distinct from the cyclical, China-driven demand fluctuations described in Section 4.

5.1 Copper Intensity by Technology

Technology	Copper Intensity	Comparison Baseline	Demand Growth Implication
Battery electric vehicle (BEV)	~60-85 kg/vehicle	vs ~20-25 kg for internal combustion engine vehicle	Roughly 3-4x the copper content per vehicle. EV penetration growth is a direct, quantifiable demand driver tracked via CAAM, EV-Volumes, and similar EV sales data.
Offshore wind turbine	~8-10 tonnes/MW	vs onshore wind ~3-4 tonnes/MW; natural gas plant ~1 tonne/MW	Offshore wind buildout (Europe, China, growing in the US) is disproportionately copper-intensive due to subsea cabling requirements.
Onshore wind turbine	~3-4 tonnes/MW	vs coal/gas generation ~1 tonne/MW	Significant but smaller multiple than offshore. Generator windings and grid connection cabling are the primary copper uses.
Solar photovoltaic	~4-5 tonnes/MW	vs traditional generation ~1 tonne/MW	Inverters, wiring, and grid connection infrastructure. China, US, and India solar additions are the primary tracked variables.
EV charging infrastructure	Variable, substantial per unit	Incremental to vehicle and grid copper demand	Public charging network buildout is an additional, separately tracked demand layer beyond the vehicles themselves.

5.2 The Long-Run Supply-Demand Gap Thesis

- **The core structural argument**: major mining companies, the International Energy Agency (IEA), and industry consultancies (Wood Mackenzie, S&P; Global, CRU Group) have published analyses projecting that planned and probable mine supply additions will fall short of energy-transition-driven demand growth over the latter half of this decade and into the next, absent a sustained period of prices high enough to incentivise new project sanctioning (given the 10-20 year lead times discussed in Section 2). This "supply gap" thesis has been a recurring theme in major producer investor presentations and sell-side commodity research since approximately 2021.
- **The thesis is a long-horizon framework, not a tactical signal**: while directionally relevant for back-curve and long-term price level views, the supply-demand gap thesis operates on a multi-year horizon and should not be treated as informative for near-term spread positioning, which is dominated by the inventory, Chinese cyclical, and macro variables discussed elsewhere in this document. The thesis has also been periodically challenged by analysts noting the historical tendency for supply forecasts to understate the cumulative effect of incremental brownfield expansions, by-product credits, and new project announcements that emerge in response to

sustained high prices.

- **EV penetration rate uncertainty**: the pace of global EV adoption — subject to policy support (subsidies, mandates), battery cost trajectories, and consumer demand — is the single largest source of uncertainty in long-run copper demand projections, with material differences between IEA, BNEF, and individual automaker scenario forecasts.

6. Global Demand Beyond China

While China dominates global copper consumption, demand from other major economies remains relevant for the marginal balance and is tracked through a distinct set of indicators.

6.1 United States Demand

- **Construction and housing**: US copper demand is significantly tied to residential and commercial construction activity (wiring, plumbing, HVAC). US housing starts and building permits data are leading indicators for this demand component.
- **Grid infrastructure and reshoring-related investment**: US grid modernisation investment (supported by federal infrastructure legislation) and a wave of domestic semiconductor and battery manufacturing facility construction (driven by industrial policy initiatives) have added an incremental, policy-driven demand layer to US copper consumption since approximately 2022.
- **The COMEX price premium dynamic**: as the US imports a substantial share of its refined copper needs, the COMEX price has periodically traded at a meaningful premium to LME, reflecting both logistics costs and — at various points — anticipated or actual trade policy actions (see Section 8).

6.2 European Demand

- European copper demand is tied to manufacturing activity (Eurozone Manufacturing PMI), the pace of the European renewable energy and grid investment programme, and the automotive sector — a sector facing its own structural transition toward EV production that affects copper intensity per vehicle produced.

6.3 Other Emerging Market Demand

- India's copper demand, while currently far smaller than China's in absolute terms, is growing rapidly alongside infrastructure investment and is increasingly cited by analysts as a potential long-run demand growth region as the country pursues its own grid and renewable buildout. Southeast Asian manufacturing growth (partly driven by supply chain diversification away from China) represents an additional, smaller but growing demand pool.

7. Exchange Inventories — The Primary Tactical Signal

Visible exchange warehouse stocks across COMEX, LME, and SHFE are the most frequently cited and most immediately actionable data series for copper spread traders. Unlike the structural, slow-moving supply and demand themes discussed in earlier sections, exchange inventory data is published daily or weekly and is the primary input to near-term spread and time-spread positioning.

7.1 The Three Exchange Stock Series

Exchange	Publication	Geographic Coverage	Key Distinction	Spread Relevance
LME	Daily	Global warehousing network: Asia (notably Johor/Port Klang), Europe (Rotterdam, Antwerp), and the Americas (New Orleans)	"On-warrant" (available) vs "cancelled" (earmarked for withdrawal, no longer counted as available) stock distinction is critical — a large cancelled-stock proportion signals imminent physical withdrawal even while headline stocks appear stable	LME cash-3-month spread (the most quoted LME copper spread) directly reflects on-warrant stock tightness; large cancellations frequently precede backwardation
COMEX	Daily	US domestic warehouses	Registered vs eligible distinction, analogous to gold (Section 11 of the gold document); reported separately and used to assess delivery-month squeeze risk	Front-month COMEX spread behaviour around first notice day is sensitive to registered stock levels relative to open interest in the expiring contract
SHFE	Weekly	Chinese domestic bonded and exchange warehouses	SHFE stocks reflect onshore Chinese conditions and are influenced by capital controls and the SHFE-LME arbitrage rather than purely global supply/demand	SHFE stock draws during the Chinese restocking season (typically post-Lunar New Year) are a closely watched seasonal demand confirmation signal

7.2 Bonded Warehouse Stocks — The "Invisible" Chinese Inventory

- **What bonded stocks represent**: in addition to SHFE-registered stocks, substantial volumes of copper are held in Chinese bonded warehouses (primarily around Shanghai), which are outside the formal SHFE exchange system and not subject to import duties until withdrawn for domestic consumption. Bonded stock levels are estimated by industry data providers (such as Shanghai Metals Market, SMM) through survey-based methods rather than official exchange disclosure, making this a less precise but still closely watched data series.
- **Why bonded stocks matter for the global balance**: copper sitting in Chinese bonded warehouses is effectively withheld from the global visible-stock count while remaining available for rapid release into the domestic market when the SHFE-LME arbitrage opens. Large swings in estimated bonded stock levels can therefore explain apparent discrepancies between visible exchange stock changes and the implied global supply/demand balance.

7.3 Days of Consumption Cover — Contextualising Stock Levels

- Absolute stock levels are most meaningfully interpreted relative to the pace of global consumption — commonly expressed as "days of demand cover." Combined LME, COMEX, and SHFE visible stocks have, at various points over the past two decades, fallen to levels representing only a small number of days of global consumption, a condition widely cited by analysts as indicative of acute physical market tightness and a contributing factor to several documented sharp price rallies and front-month backwardation episodes.

8. The COMEX-LME Arbitrage and US Trade Policy

The relationship between COMEX and LME copper prices has historically been a relatively stable, freight-and-logistics-driven arbitrage relationship. This relationship has become substantially more dynamic and policy-sensitive in the period since 2024-2025, as US trade policy actions targeting copper imports introduced a new and material driver of the spread.

8.1 Historical Basis Drivers

- In normal conditions, the COMEX-LME spread (after converting units and accounting for the US import premium) reflects: ocean freight costs from major refined copper exporting regions to US ports, the cost of US import logistics and warehousing, and the relative tightness of US domestic inventories versus the global LME-reported pool.
- Because the US imports a substantial share of its refined copper consumption (domestic mine and smelter output does not fully cover US demand), COMEX has structurally traded at a modest premium to LME in most periods, reflecting this import dependency.

8.2 Trade Policy as a New Structural Driver

- **Section 232 investigations and tariff actions**: the US government has, at various points, conducted Section 232 national security investigations into copper imports, with the possibility of tariffs on imported refined copper, concentrate, or copper-containing products. The anticipation, announcement, and implementation of such measures have each, at different points, caused significant and rapid widening of the COMEX premium over LME, as US-based buyers and traders priced in the prospective cost of tariff-affected imports and, in some episodes, accelerated import volumes ahead of anticipated tariff implementation dates — a dynamic that itself temporarily distorts both the spread and reported trade flow data.
- **Why this matters specifically for spread positioning**: tariff-driven COMEX-LME basis volatility is now a distinct, policy-calendar-driven risk factor for any trader positioned in COMEX spreads or in COMEX-versus-LME relative value, separate from the underlying physical supply/demand fundamentals described elsewhere in this document. Given the speed and magnitude with which trade policy announcements have moved this relationship in the recent past, this should be treated as an actively monitored, headline-risk-sensitive variable rather than a stable structural constant.
- **Front-running and inventory distortion**: anticipation of tariff implementation has historically incentivised accelerated shipment of refined copper into US COMEX-linked warehouses ahead of a tariff's effective date, temporarily inflating COMEX stock levels in a manner that reflects policy anticipation rather than underlying US consumption demand — analysts should be cautious about interpreting COMEX stock builds during such episodes as a simple bearish demand signal.

Because the specific tariff rates, effective dates, and exemption structures are subject to ongoing policy decisions, this document deliberately does not state a specific tariff level or status as current fact. Spread traders should verify the live policy status via primary government sources before sizing positions around this variable.

9. The Forward Curve Structure — Contango, Backwardation, and Spreads

Copper's forward curve shape reflects a blend of financial carry (interest rates and storage costs, similar to gold) and genuine physical tightness signals (similar to energy and agricultural markets) — making copper spread interpretation somewhat more complex than either a pure financial asset or a pure seasonal physical commodity.

9.1 Contango vs Backwardation in Copper

- **Full-carry contango**: in a well-supplied market with ample exchange stocks, the copper forward curve trades at a contango approximating the cost of financing and storing physical metal (interest rate plus warehousing cost, conceptually similar to the gold carry framework though copper storage costs are proportionally higher relative to value than gold's, given copper's lower value density).
- **Backwardation as a physical tightness signal**: unlike gold, where backwardation is a rare stress signal, copper backwardation (nearby trading above deferred) occurs with meaningful frequency and reflects genuine near-term physical scarcity — low exchange stocks, strong immediate consumption demand, or supply disruption (a major mine strike or smelter outage). LME cash-3-month backwardation, in particular, is one of the most closely monitored signals of acute physical tightness in the global copper market and has, at various points, reached extreme levels coinciding with historically low on-warrant LME stocks.
- **The squeeze dynamic**: because exchange-deliverable stocks can, at times, represent a small number of days of global consumption and are concentrated in specific warehouse locations, situations can arise where a small number of large holders control a disproportionate share of available nearby supply, creating conditions colloquially described as a "squeeze" — most notably referenced in connection with episodes on both COMEX and LME where front-month spread behaviour diverged sharply from the rest of the curve.

9.2 Forward Curve Zones

Zone	Contracts	Primary Driver	Secondary Driver	Key Signal
Prompt	C1–C2	Exchange on-warrant stock levels (LME, COMEX); near-term physical demand pace	Mine/smelter disruption news; Chinese restocking timing	LME cash-3-month spread; COMEX registered stocks vs open interest
Near-term	C2–C6	Chinese PMI and industrial production trend; seasonal restocking (post-Lunar New Year)	TC/RC levels (concentrate market signal); scrap availability	China Manufacturing PMI; SHFE/bonded stock changes
Mid-curve	C6–C18	Global macro growth outlook; Chinese property sector trajectory; mine supply guidance from major producers	EV/renewables demand growth trajectory; USD strength	Producer quarterly production reports and guidance; IMF/World Bank growth forecasts
Back curve	C18+	Long-run supply-demand gap thesis; major project sanctioning decisions; energy transition demand scenarios	Permitting outcomes for major undeveloped deposits; long-run USD and rate environment	IEA/Wood Mackenzie/CRU long-term outlooks; major project FID announcements

9.3 Seasonal Patterns

- **Chinese Lunar New Year effect**: Chinese industrial activity and copper consumption slow markedly in the weeks around Lunar New Year (typically late January to mid-February), as factories close for the holiday period. SHFE and bonded stocks frequently build into this period and then draw down as post-holiday restocking begins — a pattern closely watched each year, with the speed and magnitude of the post-holiday restocking draw serving as an early read on the strength of that year's Chinese demand cycle.

- **Northern Hemisphere construction seasonality**: construction-related copper demand in the US and Europe exhibits some seasonal softness during winter months, though this effect is considerably less pronounced than the seasonal patterns observed in agricultural or natural-gas-adjacent energy markets, and is generally secondary to the Chinese demand cycle and macro/inventory signals already described.

Seasonal patterns in copper are real but secondary to the China cycle, inventory signals, and macro data described elsewhere in this document. They should be used as a minor contextual overlay, not a primary trading signal.

10. Macro and Cross-Market Drivers

10.1 "Dr. Copper" — The Industrial Activity Proxy

- Copper's broad use across construction, manufacturing, power infrastructure, and consumer durables has historically given its price a documented correlation with global industrial activity and, by extension, global GDP growth expectations — the basis for the informal "Dr. Copper" characterisation as a leading indicator of economic health. This relationship is most useful as a directional, medium-term framework and should be treated with appropriate caution: periods exist (such as episodes dominated by a specific supply disruption or a Chinese stimulus announcement) where copper diverges meaningfully from broader global growth indicators for extended periods.

10.2 The US Dollar

- As with most globally traded, USD-denominated commodities, copper exhibits a generally negative correlation with the US Dollar Index (DXY): USD strength makes dollar-priced copper more expensive for buyers transacting in other currencies, weighing on demand and price at the margin, while USD weakness has the opposite effect. This relationship is generally less dominant for copper than for gold, where the USD/real-rate framework is the primary valuation driver — for copper, physical supply/demand and Chinese-specific variables frequently override the USD relationship over any given period.

10.3 Interest Rates and Global Growth Expectations

- Global monetary policy affects copper primarily through its influence on growth expectations and financial conditions for industrial and construction activity, rather than through the direct opportunity-cost-of-holding-a-non-yielding-asset channel that dominates gold pricing. Tighter global monetary policy that is expected to slow growth is generally a headwind for copper demand expectations; expectations of monetary easing in response to growth concerns can be supportive, particularly when accompanied by fiscal or infrastructure stimulus expectations, especially from China.

10.4 Related Base Metals and Cross-Commodity Signals

- **Aluminium, zinc, nickel**: while each base metal has distinct supply/demand fundamentals, broad base metals complex moves (driven by shared exposure to Chinese demand and global growth sentiment) frequently move together, and divergences between copper and the broader base metals complex (for example, copper outperforming on a specific supply disruption while aluminium remains range-bound) can be informative for distinguishing copper-specific fundamentals from broad macro/China sentiment.
- **Iron ore and steel**: while not direct substitutes, iron ore and steel prices are also heavily exposed to Chinese construction and infrastructure activity, making them a useful cross-check on the China demand narrative embedded in copper pricing at any given time.

10.5 Speculative Positioning — COT Data

- **CFTC COT for COMEX Copper**: weekly managed money net positioning data is closely tracked, as with other actively speculated metals and energy markets. Extreme net long or net short positioning is generally interpreted as a crowding/reversal-risk signal rather than a fundamental directional indicator, consistent with the positioning framework described for other markets in this document series.
- **Producer and merchant hedging**: mining companies and physical merchants hedge forward production and inventory exposure using COMEX and LME instruments; the scale and direction of this commercial hedging activity provides context for the balance of speculative positioning reflected in

the managed money category.

11. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Release Timing	Primary HG Curve Impact
LME daily warehouse stock report (on-warrant / cancelled)	London Metal Exchange	Daily	After LME close	Prompt C1–C2 — the most immediate global physical tightness signal; cancellation share is key
COMEX warehouse stocks (registered/eligible)	CME Group	Daily	After close	Front contract — delivery squeeze risk assessment around first notice day
SHFE weekly warehouse stocks	Shanghai Futures Exchange	Weekly	Friday	C2–C4 — Chinese domestic physical balance; restocking season confirmation
SMM (Shanghai Metals Market) bonded stock estimates	Shanghai Metals Market	Weekly	Variable	C2–C4 — supplements SHFE data with the "invisible" bonded inventory estimate
China Manufacturing PMI (official + Caixin)	China NBS / Caixin-S&P; Global	Monthly	1st–5th of month	C2–C8 — leading indicator for Chinese industrial copper demand
China industrial production / fixed asset investment data	China NBS	Monthly	~Mid-month	C3–C8 — confirms or revises the property/infrastructure demand trajectory
State Grid Corporation of China annual capex announcement	SGCC	Annual (with periodic updates)	Typically Q1	C6–C18 — single largest demand-side data point in global copper market
China Association of Automobile Manufacturers (CAAM) EV data	CAAM	Monthly	~Early month	C4–C12 — EV production/sales as a direct, quantifiable demand growth signal
TC/RC benchmark and spot settlements	Industry negotiation / Fastmarkets / Argus	Annual benchmark + continuous spot	Variable	C3–C12 — concentrate market tightness signal; smelter margin and curtailment risk
Major producer quarterly production reports	Codelco, Freeport-McMoRan, BHP, Antofagasta, Glencore, Southern Copper, others	Quarterly	Varies by company	C3–C18 — mine-level guidance, disruption disclosure, and production outlook revisions
ICSG (International Copper Study Group) statistical bulletin	International Copper Study Group	Monthly + annual forecast updates	Variable	All contracts — comprehensive global mine, refined, and consumption balance
IEA / Wood Mackenzie / CRU long-term copper market outlooks	IEA / Wood Mackenzie / CRU Group	Annual / periodic	Variable	C18+ — long-run supply-demand gap thesis; energy transition demand scenarios
CFTC Commitments of Traders (COMEX Copper)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — all contracts
Chilean export/production data (Cochilco)	Cochilco (Chilean Copper Commission)	Monthly	Variable	C1–C6 — the single most important country-level production data series
US housing starts / building permits	US Census Bureau	Monthly	~Mid-month	C3–C8 — US construction demand proxy

COMEX Copper sits at the intersection of long-cycle, capital-intensive mine supply economics; a demand base overwhelmingly concentrated in a single country, China; a structural, multi-year energy transition demand narrative; and an increasingly policy-sensitive cross-exchange arbitrage between COMEX, LME, and SHFE. No single framework — financial-asset-style carry, agricultural-style seasonality, or pure industrial-cycle analysis — fully captures copper's behaviour on its own; the skill in copper spread trading lies in correctly weighting the tactical inventory and Chinese cyclical signals that dominate near-term price action against the slow-moving mine supply and energy transition themes that anchor the multi-year price level.

For spread traders, the most actionable near-term signals are the daily LME on-warrant/cancelled stock dynamic, the Chinese post-Lunar New Year restocking pace, TC/RC levels as a concentrate-market tightness proxy, and — in the current environment — the policy-driven COMEX-LME basis arising from US trade measures, which has become a distinct and actively monitored risk factor separate from the underlying physical fundamentals.